

Class Exercise 12: Statistical Reporting & Graphing Figure

Instructions: **Report** the statistical results of our study and **create** a visual graph to represent the average mean results.

Requirements: To successfully complete this exercise, you are required to:

1. **draft** you're the primary findings of your results section using the results you reported in CE 11.
2. **create** 3 visual representations of the data result pattern(s) in Excel

Please use a **word processing program** (such as MS Word) to complete this exercise.

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Below is the example from the PPT that you can use as a template

Primary Analysis: The Effect of Commission vs. Omission and Apology vs. No Apology on Forgiveness

Next, I examined differences in forgiveness between the commission and omission transgression, and whether apologies functioned differently for the commission and omission transgression. To test this, I conducted a 2 (offense type: commission vs. omission) x 2 (apology condition: apology vs. no apology) between-subjects ANOVA. Consistent with hypothesis 2, there was a statistically significant main effect of offense type on forgiveness, $F(1, 591) = 23.75$, $p < .001$, $\eta^2 < .039$ (see Figure 1). Participants in the commission condition ($M = 3.79$, $SD = 1.38$) reported lower levels of forgiveness compared to participants in the omission condition ($M = 4.30$, $SD = 1.36$). Additionally, consistent with hypothesis 2, there was a statistically significant main effect of apology condition on forgiveness, $F(1, 591) = 10.48$, $p = .001$, η^2

$= .017$, $F(1, 58) = 12.22$, $p < .001$, $\eta_p^2 = .041$ (see Figure 2). Participants who received an apology ($M = 4.22$, $SD = 1.38$) than participants who did not receive an apology ($M = 3.88$, $SD = 1.36$). Lastly, contrary to hypothesis 3, there was not a statistically significant interaction between offense type and apology condition on forgiveness, $F(1, 591) = 1.14$, $p = .287$, $\eta_p^2 = .002$ (see Figure 3). This suggests that apologies promoted forgiveness to a similar extent for both commissions and omissions.

Write a rough draft of your primary results below:

Next, I examined differences in forgiveness between post-transgressional responses, and whether apologies functioned differently for levels of intellectual humility. To test this, I conducted a 2 (offense type: high intellectual humility vs. low intellectual humility) x 2 (post-transgressional response: forgiving vs. not forgiving) between-subjects ANOVA. Inconsistent with my hypothesis, there was no statistically significant main effect of intellectual humility on apology quality, $F(1,103) = 0.61$, $p = .815$. Participants in the intellectual humility condition ($M = 5.081$, $SD = 0.942$) reported no difference in levels of apology quality compared to participants in the control condition ($M = 4.931$, $SD = 1.107$). Additionally, inconsistent with our second hypothesis, there was no statistically significant main effect of post-transgressional response on apology quality, $F(1,103) = 0.22$, $p = .644$. Participants who received a forgiving post-transgressional response ($M = 4.94$, $SD = 0.918$) than participants who did not receive a forgiving response ($M = 5.039$, $SD = 1.181$). Lastly, contrary to our third hypothesis, there was not a statistically significant interaction between intellectual humility and post-transgressional response on apology quality ($F(1,103) = 0.61$, $p = .815$). This suggests that both intellectual humility and post-transgressional response are not statistically significantly related to apology quality both independently and when measured together.

Create Figures

Please follow along in class to create a figure for the main effect of message framing. Next, create a graph for the main effect of recall type. Lastly, create a graph for the interaction – try creating one on your own, or follow the steps below:

1. Create a 2x2 table of the means like the one we did in CE 7 (interactions)
 - a. *Highlight 2x2 Mean results*
 - b. *While highlighted: click “insert” tab*
 - c. *Select the 2D Column*
2. Add your **Standard Error** bars (SE) to the Figure
 - i. First, add the SEs anywhere in the Excel spreadsheet, as long as they follow the same 2X2 format. For example...

	Positive	Negative
Unity	9.32	9.07
Journey	9.44	9.30
SE	0.20	0.20
	0.21	0.21

- ii. Highlight the first color bars (only one bar)

1. Chart Design -> Add Chart Element -> Error Bars -> More Error Bar Options
 - a. Direction (both, End Style (Cap), Error Amount (Custom/Specify Value)
 - b. Positive error value: (highlight your SE values for IV 1 levels OR type in the value)
 - c. Negative error value: highlight your SE values for IV 1 levels OR type in the value (same values as the positive)

***If you highlighted two conditions (the same color but across two conditions, you can tell Excel to include both error bars by adding a comma “,” in between the values

Example: in the positive error value, highlight the SE value for IV 1 level across the one condition of IV 2, **add comma**, then highlight the SE value for IV 1 level for the second condition of IV 2

[0.03,0.04]

iii. Highlight the second color bars (only one bar)

1. Chart Design -> Add Chart Element -> Error Bars -> More Error Bar Options

a. Click the statistical illustration (i.e., the bars)

i. Direction: click Both,

ii. End Style: click Cap

iii. Error Amount: click Custom, then Specify Value

b. Positive error value: highlight your SE values for the second color bar OR type in value

c. Negative error value: highlight your SE values for the second color bar OR type in value (same values as the positive)

3. Y-axis Title:

a. Click “Chart Design”

i. Add Chart Element -> Axis Titles -> Primary Vertical

ii. Write in your Y-axis title (Dependent Variable)

4. Y-axis Values:

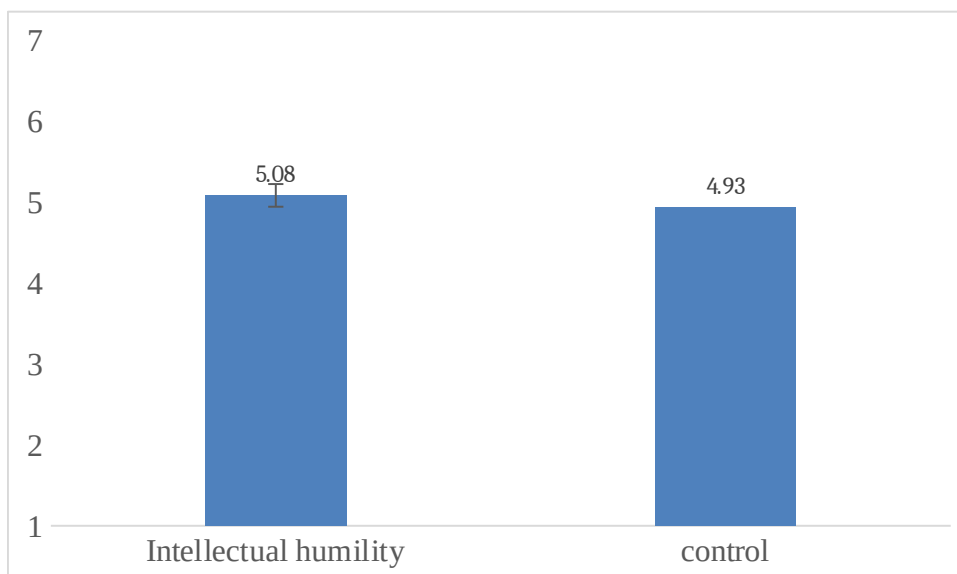
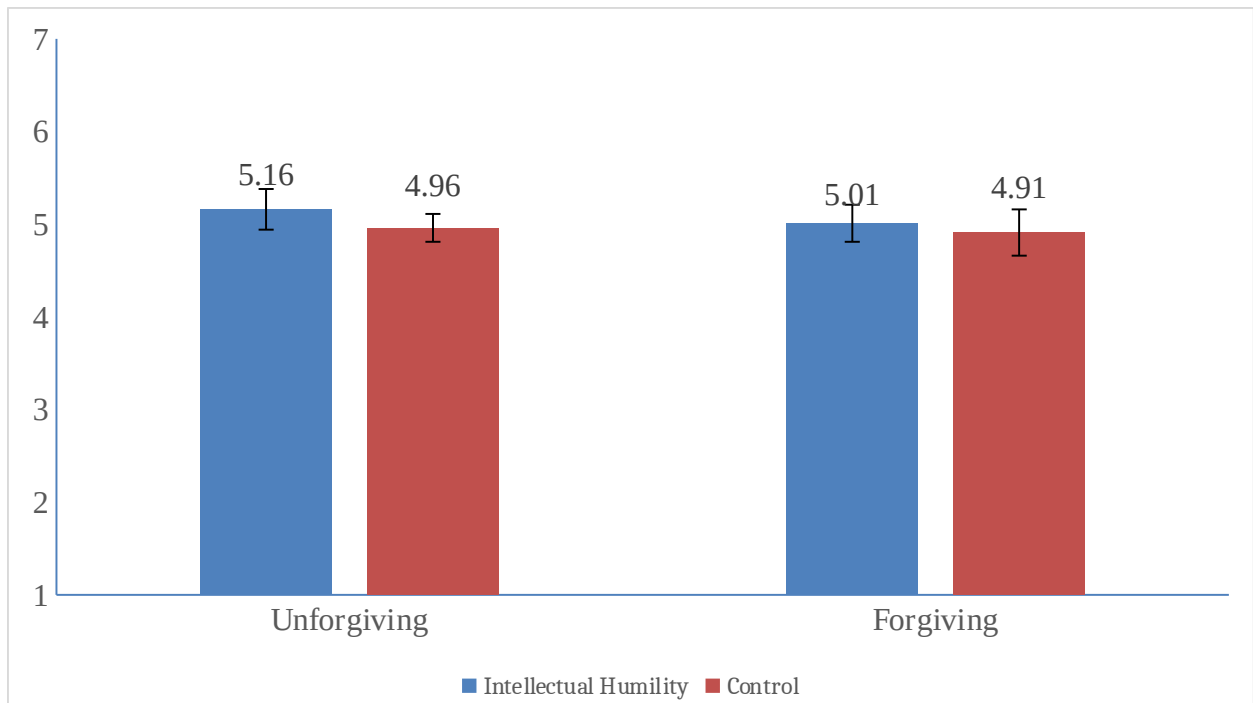
a. Double click Y-axis values

i. Click “Text Options” in the Format Axis options

ii. Edit the units, minimum & maximum bounds to your desire (roughly 6 values to illustrate the Y-axis values is the norm)

5. Adjust Figure font sizes and colors to your preference

6. Finally, copy and paste your Figure to this document





APA Checklist & Tips!

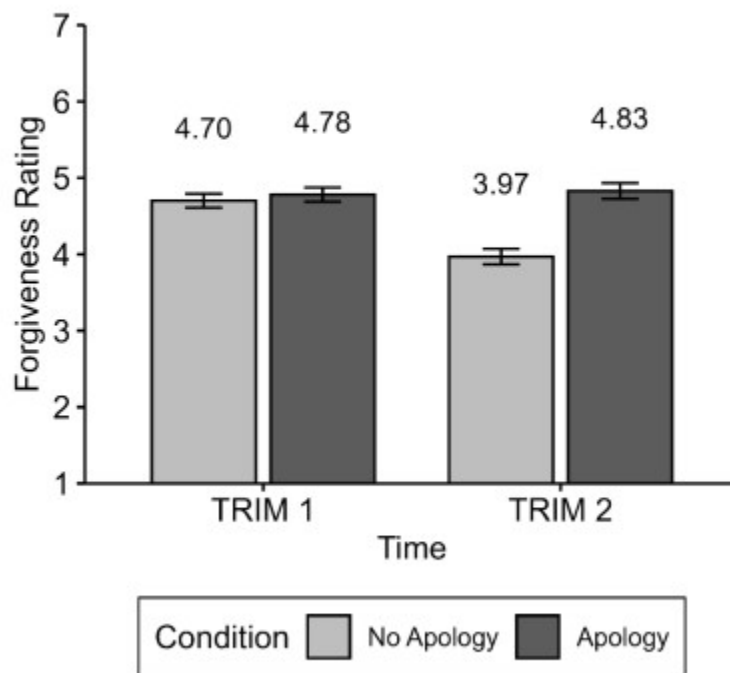
- Use colors that are easy to read.
- Ensure all text in the graph is in **Times New Roman, 12pt font**.
- Add **x-axis and y-axis lines**, ensuring the **y-axis reflects the scale range** for the dependent variable (DV).
- **Add labels** to the x-axis and y-axis.
- Include **data labels** on the outside end of each bar to display the means.

- Make all **text and lines black** for consistency.
- The background lines may be removed, but it's okay to keep them if necessary.
- Include a **note** at the bottom stating that the error bars represent standard errors.
- Ensure the **figure # and title** are in APA format (see example below)

Example Figure

Figure 1

The Effect of Apology Condition on Forgiveness from Time 1 to Time 2, Study 1



Note. Error bars represent standard errors.